

**Master of Computer Application (M.C.A.) (Two Year Programme) -
2nd Sem. (Credit Based Evaluation and Grading System)**

(Batch 2024-26)

(2525)

Paper: CSL-4097 Deep Learning

Time allowed: 3 hrs.

Max. Marks: 70

**Note : Attempt any five questions by selecting at least one question from each section.
All questions carry equal marks.**

Section A.

1. What is a biological neuron? Compare it with artificial neuron. Also explain working of McCulloch-pitts model of neuron.
2. What is a feedforward neural network? Give some applications of feedforward networks.

Section B

3. What is meant by Deep learning? Explain the challenge of vanishing and exploding gradients in deep learning models and the method to overcome the challenges.
4. Define term "Overfitting". What are different methods of avoiding overfitting thro' regularization?

Section C

5. What are main characteristics of Recurrent neural networks? Write a note on Bidirectional RNNs.
6. Give the architecture of CNN. Give salient features of AlexNet and LeNet

Section D

7. What are generative models? How are Generative Adversarial Networks used in AI?
8. Write a note on multi-view deep learning.

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